Lateral-Acceleration-Based Torque Tracking Control for Column Electric Power Steering under Varying Road Friction Conditions

Abstract

Conventional assist maps in column electric power steering (C-EPS) are calibrated for dry roads and do not directly regulate steering torque relative to a specified reference. When tire–road friction changes, the road-reaction torque changes, and the steering effort produced by the calibrated map departs from its intended value. This study examines whether torque-error feedback can be added to such a map without recalibrating it. Three structures are compared: a lead–lag-compensated assist map (Map), direct proportional–integral–derivative torque feedback (PID), and a plug-in combination (MapPID) in which the calibrated map and its compensator are kept unchanged and only a parallel PID branch is tuned. The desired torsion-bar torque is defined from measured vehicle speed and lateral acceleration, without online estimation of tire–road friction or rack force. A local steady-state analysis shows that a static assist map attenuates road-reaction variations only by a finite factor set by its slope, whereas an added integral branch removes the steady-state error for any map calibration while the actuator is unsaturated; the feedback branch then supplies only the residual that the map leaves. The controllers are assessed on a nonlinear C-EPS–vehicle model with tire-force saturation, slip-dependent pneumatic trail, motor dynamics, bounded assistance, and quantized sensor noise, under common feasibility constraints, Pareto-based calibration, and independent dry-cornering, speed-ramp, friction-change, edge, and weave scenarios.

Keywords: Electric power steering; Assist map; Torque tracking; Lateral acceleration; Road friction; PID control.

1. Introduction

Electric power steering (EPS) assists the driver with an electric motor instead of a hydraulic pump, which improves efficiency and makes the steering characteristic adjustable in software [1]. In column-type EPS (C-EPS), the motor acts on the steering column. The controller usually reads the motor assistance from a static, nonlinear assist map indexed by the measured torsion-bar torque and the vehicle speed, followed by a lead–lag compensator that stabilizes the column and limits noise amplification [1,2]. The map is calibrated on a dry road so that the torque felt by the driver follows a desired steering-effort characteristic.

This calibration holds only for the condition in which it was made. When tire–road friction decreases, the front tire saturates earlier and its pneumatic trail shrinks, so the road-reaction torque returned to the column falls [3,5,12]. The map reacts only to the measured torque and has no comparison with a reference, so the steering torque settles below the desired value: the change is reduced only by the factor $1/(1 + s_m)$ of the local map slope s_m (Section 3.2), and the slope is limited by stability and noise [1,9]. The result is over-assist, a lighter steering wheel with reduced road feedback [3]. Whether the deviation is perceptible depends on the maneuver, since drivers discriminate steady torque differences of about 10–18% [18]; it matters mainly in transients and near the assist limit. Recalibrating the map for every condition is costly because the map and the stabilizing compensator are coupled [2,9].

Existing studies address this in three ways. Road-adaptive strategies identify the road with neural networks [5], compensate the assist current with an estimated friction coefficient [4,6], or select among multiple maps by adhesion and axle load [3]; they need friction or axle-load estimates that are not measured directly. Fuzzy [10] and adaptive neuro-fuzzy [11] controllers address EPS nonlinearities and current limits but not this deviation. Zhao et al. [7] combined a characteristic curve with a PID loop on the motor torque, but the loop tracks the torque requested by the map and does not treat friction changes. Lee et al. [8,9] replaced the map by tracking a target steering torque, with an adaptive sliding-mode controller [8] or a lead–lag controller with a disturbance observer built on an estimated rack force [9]; their vehicle models and tests do not include friction transitions with nonlinear tire saturation, and the calibrated map is discarded. None of the reviewed studies keeps an existing calibrated map and adds torque feedback to it.

This study therefore asks what a feedback branch adds to a calibrated map and what the map adds to direct feedback. Three structures share one torque reference defined from

measured vehicle speed and lateral acceleration [5], which needs no friction estimate: the assist map with lead–lag compensation (Map), PID torque feedback (PID), and a plug-in combination (MapPID) that keeps the Map design unchanged and tunes only a parallel PID branch. They are compared on a nonlinear C-EPS–vehicle model with tire saturation, motor dynamics, and noisy sensors, under common stability constraints.

The main contributions of this study are as follows:

- A steady-state analysis showing that the assist map leaves a residual torque error set by its slope, which integral feedback removes for any map calibration while the actuator is unsaturated.
- A plug-in MapPID structure that retains the calibrated map and tunes only the PID branch, with an anti-windup rule adapted to the combined forward gain.
- A common simulation protocol with Pareto-based calibration and independent dry, speed-ramp, friction-change, edge, and weave scenarios, which quantifies torque tracking, high-frequency assist-torque content, and reverse assistance.

2. System Modeling and Problem Formulation

The plant couples three subsystems: the steering column with the assist motor, the front and rear tires, and the lateral and yaw motion of a single-track vehicle body. Figure 1 shows how they are connected. The column delivers the lower-column angle θ_2 to the tires; the tires return the lateral forces to the vehicle body and the road-reaction torque T_r to the column; the body returns the sideslip angle β and the yaw rate γ to the tire slip angles. This loop is the path through which a change of tire–road friction μ changes the torque felt at the steering wheel. The external inputs are the driver steering-wheel angle θ_1, the assist torque T_a, the vehicle speed v, and μ. Sections 2.1–2.3 model each subsystem, Section 2.4 combines them, and Sections 2.5–2.6 define the torque reference and the tracking problem shared by all controllers.

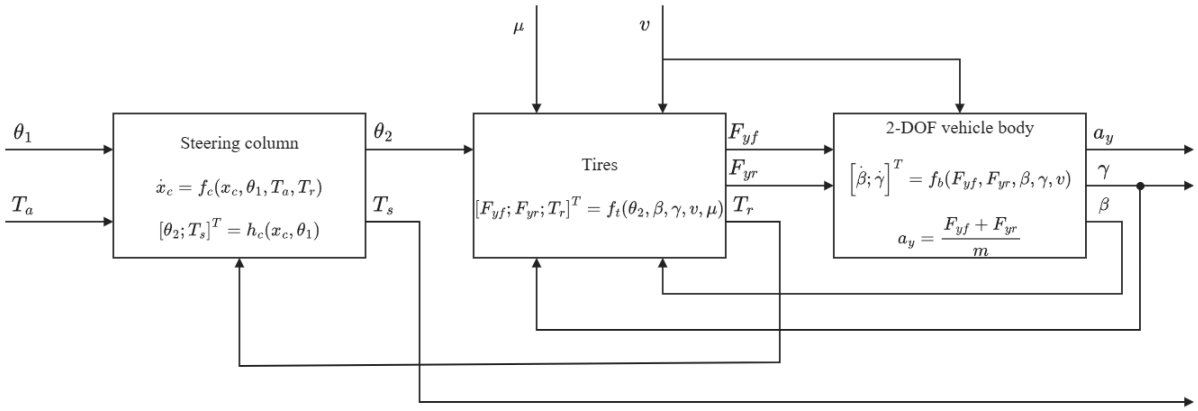


Fig. 1. Block diagram of the coupled C-EPS–vehicle plant: steering column, tires, and two-degree-of-freedom vehicle body.

2.1. Steering Column and Assist Motor

The mechanical structure of the column-type EPS is shown in Fig. 2. The upper column, driven by the steering wheel, is connected to the lower column through a torsion bar of stiffness K, which also serves as the torque sensor. The motor applies the assist torque T_a to the lower column through a reduction gear, and the lower column drives the rack through the steering gear of ratio n_{st}. The steering-wheel angle θ_1 is prescribed as the driver input. The lower-column dynamics and the torsion-bar torque are modeled as [1]

$$J_{col}\ddot{\theta}_2 = T_s + T_a - T_r - C_{col}\dot{\theta}_2 - T_f \tanh(c\dot{\theta}_2) \tag{1}$$

$$T_s = K(\theta_1 - \theta_2) \tag{2}$$

where θ_2 is the lower-column angle; J_{col} and C_{col} are the equivalent inertia and viscous damping of the lower column, including the motor and gear referred to the column; and $T_f \tanh(c\dot{\theta}_2)$ is a smooth approximation of Coulomb friction with amplitude T_f. Both T_a and the road-reaction torque T_r are referred to the column side of the mechanism.

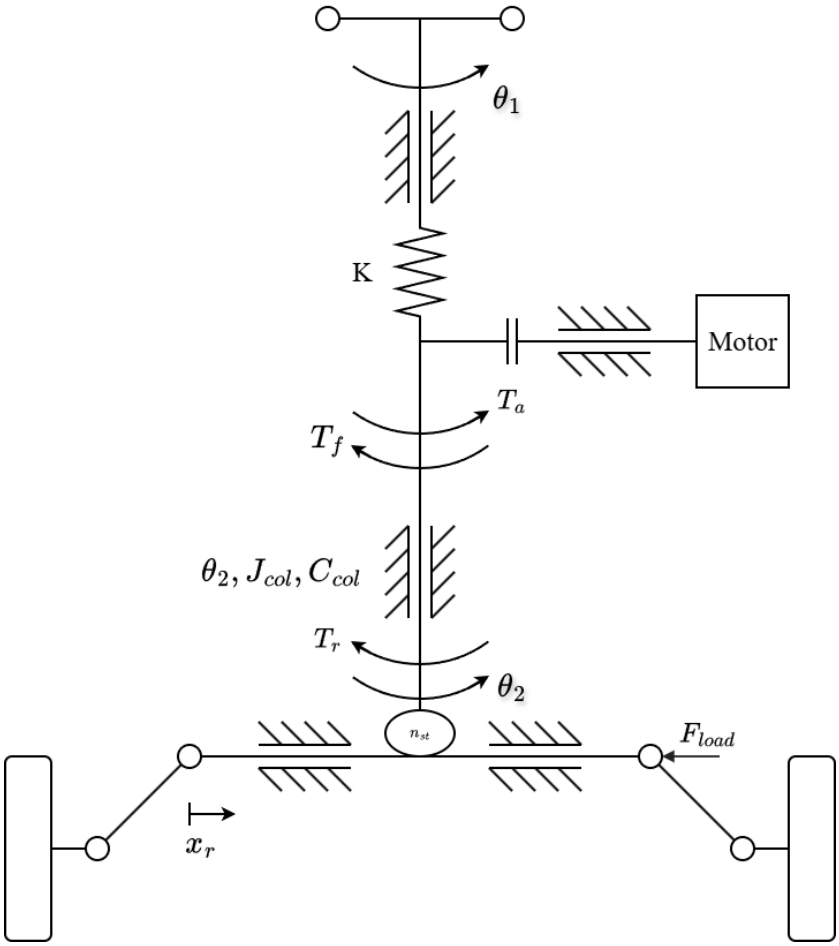


Fig. 2. Mechanical structure of the column-type EPS.

Prescribing $\theta_1(t)$ allows T_s to change when the road reaction or the motor assistance changes. If the driver torque were imposed instead, it would fix the sensed torque at steady state and hide the over-assist mechanism examined here. The prescribed-angle driver therefore supports a controlled comparison of steering-effort responses, but it does not represent the driver's neuromuscular response or the correction of steering angle in response to changing effort.

2.2. Tire Model and Road-Reaction Torque

Figure 3 defines the road-plane quantities of the front tire. The front-wheel angle is approximated by $\delta_f = \theta_2/n_{st}$. Under the small-angle and constant-longitudinal-speed assumptions, the front and rear slip angles are [13]

$$\alpha_f = \frac{\theta_2}{n_{st}} - \beta - \frac{l_f \gamma}{v} \tag{3}$$

$$\alpha_r = -\beta + \frac{l_r \gamma}{v} \tag{4}$$

where β is the vehicle sideslip angle, γ is the yaw rate, and l_f and l_r are the distances from the center of gravity to the front and rear axles. Longitudinal and lateral load transfer are neglected, giving the static axle loads

$$F_{zf} = mg \frac{l_r}{l_f + l_r} \tag{5}$$

$$F_{zr} = mg \frac{l_f}{l_f + l_r} \tag{6}$$

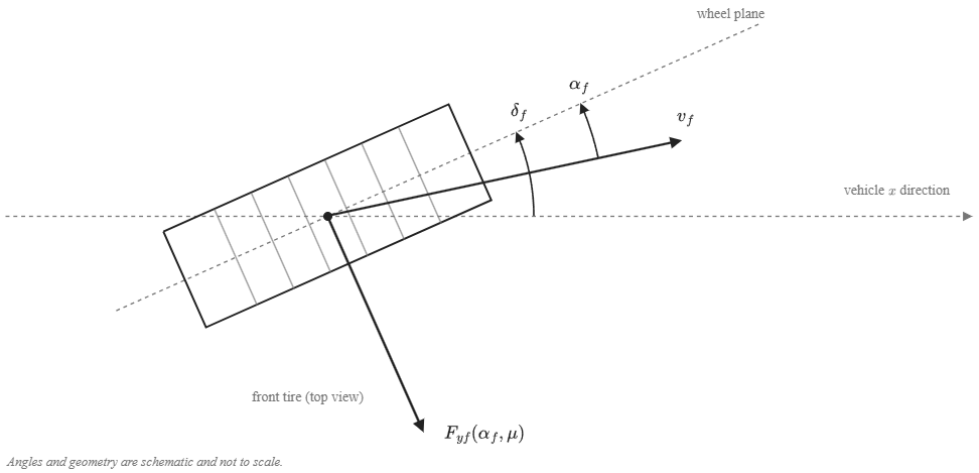


Fig. 3. Top-view geometry of the front-tire steering angle, slip angle, and lateral force.

Each axle is represented by one equivalent tire. To retain force saturation when road friction changes, the lateral forces are calculated with a simplified Pacejka Magic Formula [12]:

$$F_{yi} = D_i \sin[C \tan^{-1}(B_i \alpha_i - E(B_i \alpha_i - \tan^{-1}(B_i \alpha_i)))], \quad i \in \{f, r\} \tag{7}$$

$$D_i = \mu F_{zi}, \quad B_i = \frac{C_{\alpha i}}{C D_i} \tag{8}$$

where μ is the common tire–road friction coefficient, $C_{\alpha i}$ is the axle cornering stiffness, and C and E are shape and curvature factors. This parameterization preserves the initial slope $C_{\alpha i}$ while reducing the peak force D_i as μ decreases, so the tire saturates earlier on a lower-friction surface.

The front lateral force acts on the steering axis through an effective moment arm that contains a constant caster component and a slip-dependent pneumatic component. Adapting the pneumatic-trail relation used in [3], the model defines

$$t_p = \max(0, t_0 - \frac{t_0 C_{\alpha f} |\tan\alpha_f|}{3\mu F_{zf}}) \tag{9}$$

$$e_{eff} = e_c + t_p, \quad e_c = e_{p0} - t_0 \tag{10}$$

where e_{p0} is the effective moment arm at zero slip, t_0 is the zero-slip pneumatic trail, and e_c is the constant caster component. The nonnegative bound on t_p represents the loss of pneumatic trail as the modeled contact patch reaches full sliding; retaining e_c prevents the total moment arm from being forced to zero. These bounds are modeling choices added to the source relation and do not reproduce the complete aligning-moment behavior of a measured tire. The road reaction referred to the steering column is

$$T_r = \frac{e_{eff}}{n_{st}} F_{yf} \tag{11}$$

Equations (8)–(11) contain the two effects of friction on the column: a lower μ reduces the attainable F_{yf} and, at the same slip angle, shortens the pneumatic trail t_p. Both reduce T_r.

2.3. Single-Track Vehicle Body

The lateral and yaw motion of the vehicle is governed by the nonlinear single-track balances [13]:

$$a_y = \frac{F_{yf} + F_{yr}}{m} \tag{12}$$

$$\dot{\beta} = \frac{a_y}{v} - \gamma \tag{13}$$

$$\dot{\gamma} = \frac{l_f F_{yf} - l_r F_{yr}}{I_z} \tag{14}$$

where m and I_z are the vehicle mass and yaw moment of inertia. The lateral acceleration a_y of Eq. (12) is also the measured signal used by the torque reference in Section 2.5.

2.4. Coupled Plant and Over-Assist Mechanism

Combining Eqs. (1)–(14), the coupled steering-column and vehicle plant of Fig. 1 has four states, θ_2, $\dot{\theta}_2$, β, and γ; the motor, compensator, and controller states are added separately in the closed-loop simulations. For a stationary steering column, Eq. (1) becomes

$$T_s = T_r - T_a \tag{15}$$

because the smoothed Coulomb-friction term vanishes at $\dot{\theta}_2 = 0$. Equations (7)–(15) expose the mechanism of interest: decreasing μ lowers the tire-force capacity and the pneumatic trail and therefore T_r. If the assistance does not decrease by the same amount, Eq. (15) predicts a reduction of the steering torque transmitted through the torsion bar, that is, over-assist.

2.5. Lateral-Acceleration-Based Torque Reference

The desired steering torque is obtained from the speed- and lateral-acceleration-dependent steering-effort data reported by Li and Xia [5]. The four lateral-acceleration levels published in that study are reproduced in Table 1 and are treated here as a reference characteristic rather than as a universal driver-preference model.

Table 1. Ideal steering-wheel torque reference (N·m), adopted from [5].

Lateral acceleration	20 km/h	40 km/h	60 km/h	80 km/h	100 km/h
0.1 g	1.0	1.7	2.1	2.3	2.6
0.2 g	1.5	2.0	2.4	2.8	3.0
0.3 g	1.6	2.3	2.8	3.1	3.3
0.4 g	1.8	2.4	2.9	3.3	3.4

A zero-torque point is added at zero lateral acceleration. Monotone piecewise cubic Hermite interpolation is applied first along lateral acceleration and then along vehicle speed, after which the result is sampled on a finer table for implementation. Odd symmetry supplies the negative-turning branch, while inputs beyond 20–100 km/h or 0–0.4 g are clipped to the nearest boundary:

$$T_{ref}(v, a_y) = \mathrm{sgn}(a_y)\,\mathrm{widehat}T(v, |a_y|) \tag{16}$$

$$\mathrm{widehat}T(v, 0) = 0 \tag{17}$$

where widehat$T$ denotes the interpolated positive-acceleration characteristic. Adding the zero point, imposing odd symmetry, selecting PCHIP interpolation, and clipping outside the published grid are implementation choices of this study. They make the reference continuous through straight-ahead driving without extrapolating unreported steering-effort values.

The reference uses measured vehicle speed and lateral acceleration and does not require an estimate of μ or rack force. Because a_y is an input to the reference, a friction reduction can also change the target by changing the vehicle response. The controller therefore regulates effort relative to the motion actually produced by the vehicle instead of enforcing the same torque at a fixed steering-wheel angle on every surface.

2.6. Torque-Tracking Objective

The common control objective is to regulate the measured torsion-bar torque toward the reference:

$$e_T = T_s - T_{ref}\left(v, a_y\right) \tag{18}$$

For T_{ref}ne0, $\mathrm{sgn}\left(T_{ref}\right)e_T < 0$ denotes a torque magnitude below the target in the commanded turning direction. Reversal intervals are evaluated with magnitude-based measures because the signed interpretation becomes ambiguous near a zero crossing. All controllers act through the same bounded assist-torque channel:

$$|T_a| \le T_{a,max}(v) \tag{19}$$

where $T_{a,max}(v)$ is derived from the modeled dry-road assistance requirement and is specified with the actuator implementation. The comparison therefore asks how each controller reduces the torque-tracking error under varying friction while satisfying the same motor dynamics, assist limit, sensor conditions, and stability requirements.

3. Control Strategy and Assist-Map Limitation

This section presents the three control structures: conventional assist-map control with lead–lag compensation (Map), direct PID steering-torque feedback (PID), and the plug-in combination (MapPID), in which the Map design is retained unchanged and a PID branch is added in parallel. A steady-state sensitivity analysis quantifies the limitation of a static assist map when tire–road friction changes and shows why an added integral branch removes it independently of the map calibration.

3.1. Conventional Assist-Map Control with Lead–Lag Compensation

The conventional controller follows the standard automotive architecture in which basic motor assistance is scheduled through a static two-dimensional map from measured torsion-bar torque and vehicle speed, followed by dynamic stabilizing compensation [1]. Under nominal dry-road conditions ($\mu = 0.8$), the required assistance at each calibration breakpoint is determined from the static torque equilibrium of Eq. (15):

$$T_{s,cal} = T_{ref}\left(v, a_y\right) \tag{20}$$

$$T_{a,cal} = T_r\left(v, a_y, 0.8\right) - T_{ref}\left(v, a_y\right) \tag{21}$$

The nominal dry-road assist map is calibrated across 17 vehicle speeds from 20 to 100 km/h in increments of 5 km/h. At each speed, feasible lateral accelerations are evaluated in increments of 0.005 g. The raw assistance is bounded by a symmetric torque dead zone $T_{s0} = 0.3$ N·m around straight-ahead driving to prevent motor chattering from sensor noise, and by a maximum local slope K_{max}. At each calibration breakpoint, the allowable assist increment is constrained by $\bar{M}_k \le \bar{M}_{k-1} + K_{max}(|T_s|_k - |T_s|_{k-1})$, preserving nondecreasing assistance while preventing excessively steep transitions:

$$M(v, T_s) = \mathrm{sgn}(T_s)\bar{M}(v, |T_s|) \tag{22}$$

where $\bar{M}(v, |T_s|)$ is the bounded monotonic assist characteristic for positive torque. The resulting characteristic is tabulated over measured steering torque from 0 to 5 N·m with linear interpolation across torque and vehicle speed.

To stabilize the closed loop around the steering-column resonance (about 6.2 Hz) while limiting high-frequency noise amplification from the high assist gain, the static map output is processed by a linear lead–lag compensator $C_{map}(s)$ [1]:

$$u_{map}(s) = C_{map}(s)M\left(v_m, T_{s,m}\right) \tag{23}$$

$$C_{map}(s) = \left(\frac{\frac{s}{z}+1}{\frac{s}{p}+1}\right)\left(\frac{\frac{s}{z_1}+1}{\frac{s}{p_1}+1}\right) \tag{24}$$

where z and p are the zero and pole frequencies of the lead stage, and z_1 and p_1 are those of the lag stage. The lead stage provides phase advance around the column resonance, while the lag stage provides high-frequency gain roll-off. Both stages have unity steady-state gain

($C_{map}(0) = 1$), so the calibrated static characteristic is preserved. The closed-loop signal flow of the assist-map controller is shown in Fig. 4.

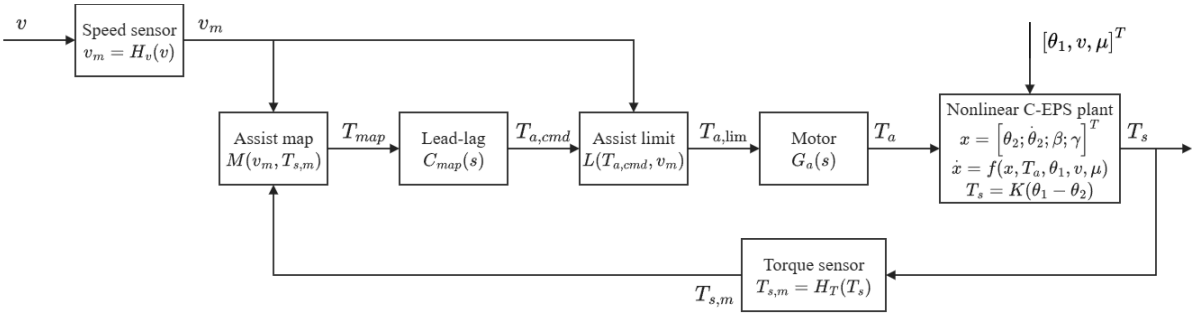


Fig. 4. Closed-loop signal flow of the assist-map controller.

3.2. Steady-State Limitation of Assist-Map Control

The limitation of the assist map under friction variation can be evaluated through a local steady-state sensitivity analysis. At a constant vehicle speed and fixed lateral acceleration, column equilibrium under assist-map control requires

$$T_s + M(v, T_s) = T_r(v, a_y, \mu) \tag{25}$$

where the Coulomb friction term vanishes at steady state ($\dot{\theta}_2 = 0$) and $C_{map}(0) = 1$. Linearizing Eq. (25) with respect to a change in road-reaction torque ΔT_r yields

$$\Delta T_s = \frac{\Delta T_r}{1 + s_m} \tag{26}$$

$$s_m = \frac{\partial M(v, T_s)}{\partial T_s} \tag{27}$$

where s_m denotes the local slope of the assist map at the operating point.

Equation (26) shows that the change in sensed torsion-bar torque is attenuated only by the factor $1/(1 + s_m)$. Because the target torque $T_{ref}(v, a_y)$ is unchanged at fixed speed and lateral acceleration, the change in tracking error equals ΔT_s:

$$\Delta e_T = \frac{\Delta T_r}{1 + s_m} \tag{28}$$

When friction drops from the dry value ($\mu = 0.8$), the road reaction T_r decreases because of tire-force saturation and pneumatic-trail reduction. Because s_m is positive and finite ($s_m \leq K_{max}$), Δe_T is nonzero and has the same sign as ΔT_r: T_s falls below the desired value,

which is the over-assist condition. Increasing K_{max} reduces Δe_T but raises the loop gain and amplifies sensor noise, and no finite slope removes Δe_T. The same slope limit also prevents the map from reaching the calibration points of Eq. (21) where the required local slope exceeds K_{max}, so a residual error can remain even on a dry road. In contrast, any controller with integral action on e_T enforces $e_T = 0$ at steady state whenever the actuator is not saturated.

3.3. PID Steering-Torque Feedback Control

The PID controller regulates the torque-tracking error $e_T = T_{s,m} - T_{ref}(v_m, a_{y,m})$ directly. To avoid a derivative kick after rapid changes of the reference, the derivative action is placed on the measured torsion-bar torque $T_{s,m}$ rather than on the error, and it is filtered by a first-order lag:

$$u_P = K_p e_T \tag{29}$$

$$\dot{u}_I = K_i e_T + K_{aw} \Delta T_a \tag{30}$$

$$u_D(s) = \left(\frac{K_d s}{T_f s + 1}\right) T_{s,m}(s) \tag{31}$$

where K_p, K_i, and K_d are the proportional, integral, and derivative gains, T_f is the derivative filter time constant, and K_{aw} is the anti-windup tracking gain.

In Eq. (31), the derivative term has a positive sign because an increasing measured torque T_s indicates insufficient assistance in the column convention of Eq. (1), which requires an increase of assist torque. The total unconstrained command is

$$T_{a,cmd} = u_P + u_I + u_D \tag{32}$$

Actuator saturation is handled by back-calculation anti-windup. The difference between the bounded motor command $T_{a,lim}$ and the raw command $T_{a,cmd}$ is fed back to the integrator in Eq. (30):

$$\Delta T_a = T_{a,lim} - T_{a,cmd} \tag{33}$$

$$K_{aw} = \frac{K_i}{K_p} \tag{34}$$

This back-calculation discharges the integrator whenever the motor saturates at $T_{a,max}(v_m)$ and allows a fast recovery when the error reverses. Figure 5 shows the signal flow of the PID torque-feedback controller.

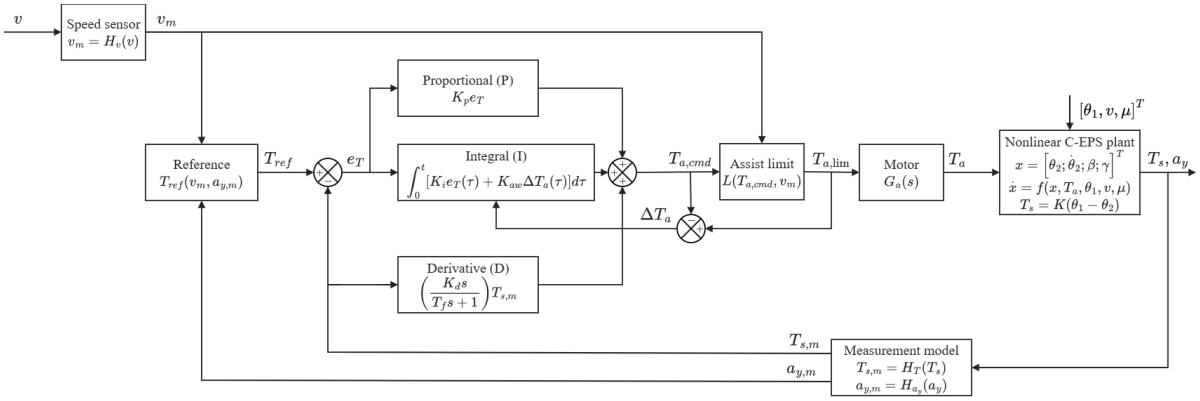


Fig. 5. Closed-loop signal flow of the PID steering-torque controller.

3.4. Plug-In Map-PID Control Structure

The MapPID structure keeps the assist map and lead–lag compensator of Section 3.1 exactly as calibrated for Map, including the slope limit K_{max}, the dead zone T_{s0}, and the compensator frequencies, and adds the PID branch of Section 3.3 in parallel (Fig. 6). Only the four branch parameters K_p, K_i, K_d, and T_f are tuned, with the map block frozen. The branch is therefore a plug-in: it can be attached to an existing calibration without changing the map that encodes the dry-road steering characteristic. The assist command is

$$T_{a,cmd} = u_{map} + u_{PID} \tag{35}$$

$$u_{PID} = u_P + u_I + u_D \tag{36}$$

where u_{map} is computed from Eq. (23) and u_{PID} from Eqs. (29)–(31).

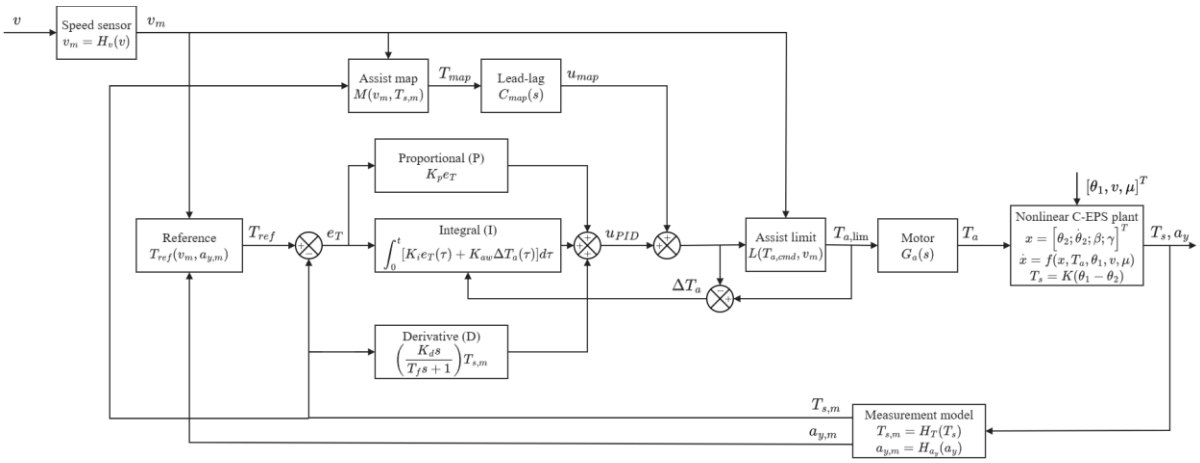


Fig. 6. Closed-loop signal flow of the plug-in MapPID controller.

The steady-state behavior of the plug-in branch follows from Eq. (25). With the branch attached, the column equilibrium becomes $T_s + M(v, T_s) + u_{PID} = T_r(v, a_y, \mu)$. While the

closed loop is stable and the actuator is not saturated, the integrator stops changing only when $e_T = 0$, so that $T_s = T_{ref}$ and the branch settles at

$$u_{PID,ss} = T_r\left(v, a_y, \mu\right) - T_{ref}\left(v, a_y\right) - M\left(v, T_{ref}\right) \tag{37}$$

Equation (37) states the two properties on which the plug-in structure relies. First, the steady-state accuracy does not depend on the map: any monotonic map, whatever its slope or calibration, leads to $e_T = 0$. Second, the branch supplies only the residual that the map leaves. At a dry-road calibration point reached by the map, Eq. (21) gives $M\left(v, T_{ref}\right) = T_r\left(v, a_y, 0.8\right) - T_{ref}$, so $u_{PID,ss} = T_r\left(v, a_y, \mu\right) - T_r\left(v, a_y, 0.8\right)$, which is zero on the dry road and equal to the friction-induced change of road reaction otherwise. Where the slope limit prevents the map from reaching its calibration point, $u_{PID,ss}$ also contains that calibration residual.

The map is not neutral for the dynamics of the branch. Around an operating point, the forward path from measured torque to assist torque is $s_m C_{map}(s) + C_{PID}(s)$, where $C_{PID}(s)$ is the linearized PID transfer function. The map slope therefore adds to the low-frequency loop gain, which can reduce the proportional gain needed by the branch, but it also adds to the loop gain near the column resonance, so the stability margins of the branch depend on the map it is attached to. For this reason, the stability requirements are verified on the combined loop, and the sensitivity of the tuned branch to the map it is plugged into is examined separately. Both branches share the common saturation limit:

$$T_{a,lim} = \text{clip}\left(T_{a,cmd}, -T_{a,max}(v_m), T_{a,max}(v_m)\right) \tag{38}$$

Because both the map, with maximum slope K_{max}, and the proportional path K_p contribute to the forward gain between sensed torque and assist torque, the effective forward gain is bounded by $K_p + K_{max}$. The back-calculation anti-windup gain is scaled accordingly:

$$K_{aw} = \frac{K_i}{K_p + K_{max}} \tag{39}$$

When $K_{max} = 0$, Eq. (39) reduces to the PID rule of Eq. (34), and when K_p is small, the map slope keeps the anti-windup gain finite. The map therefore remains responsible for the nominal dry-road assistance, while the plug-in branch corrects the deviations caused by friction changes, tire saturation, and the slope-limited calibration.

4. Simulation Setup and Evaluation Methodology

4.1. Simulation Model and Implementation

The three controllers are evaluated using the nonlinear C-EPS–vehicle model described in Section 2. The adopted steering, vehicle, and tire parameters are summarized in Table 2. Steering-column parameters are based on [1], whereas the tire and vehicle model formulations follow [12,13]. The steering ratio and trail dimensions are prescribed modeling parameters. The magnitude of the friction-induced steering-torque reduction depends on the assumed zero-slip pneumatic trail. For a friction decrease from μ = 0.8 to 0.3 under the fixed-lateral-acceleration condition considered in the sensitivity analysis, varying t_0 from 17 to 35 mm changes the predicted reduction in sensed torque from approximately 6% to 13% without motor assistance. The nominal parameter value gives a reduction of about 9–10%. The model therefore captures the expected direction of the effect, while its magnitude remains dependent on the assumed steering geometry.

Table 2. Parameters of the nonlinear C-EPS–vehicle model.

Parameter	Symbol	Value	Unit
Torsion-bar stiffness	K	143.24	N·m/rad
Lower-column inertia	J_{col}	0.11	kg·m²
Column viscous damping	C_{col}	1.35	N·m·s/rad
Coulomb friction amplitude	T_f	2	N·m
Friction smoothing coefficient	c	100	s/rad
Vehicle mass	m	1573	kg
Yaw moment of inertia	I_z	2873	kg·m²
Front axle distance	l_f	1.10	m
Rear axle distance	l_r	1.58	m
Front/rear cornering stiffness	$C_{\alpha f}, C_{\alpha r}$	160000	N/rad
Magic Formula shape factor	C	1.3	-
Magic Formula curvature factor	E	−1	-
Steering ratio	n_{st}	16	-
Initial effective moment arm	e_{p0}	0.048	m
Initial pneumatic trail	t_0	0.025	m

The model is evaluated between 20 and 100 km/h and friction coefficients from 0.2 to 0.8. Below 20 km/h, the inverse-speed terms in the vehicle equations and the quasi-static tire assumptions become less reliable. The feasible lateral-acceleration range is limited by tire friction and by the small-angle assumptions. At 20 km/h, the adopted model permits approximately 0.2 g, whereas the upper limit approaches the friction-dependent tire-force limit at higher speeds. Therefore, calibration and test points are restricted to feasible operating conditions rather than the full rectangular range of the reference table. Equal tire–road friction is assumed on the left and right sides of the vehicle, and the steering-wheel angle is prescribed independently of the driver's response to steering torque.

The motor dynamics are approximated by a first-order transfer function [1]:

$$G_m(s) = \frac{\omega_m}{s + \omega_m}, \quad \omega_m = 2\pi(100)\ \text{rad/s} \tag{40}$$

The common motor model includes a 1 ms computation delay and the speed-dependent assist limit defined in Eq. (19). The latter is calculated from the maximum required dry-road assistance:

$$T_{a,max}(v) = \max_{0 \le a_y \le a_{y,lim}(v)} \left[T_r\left(v, a_y, 0.8\right) - T_{ref}\left(v, a_y\right)\right] + T_f \tag{41}$$

$$a_{y,lim}(v) = \min\left(a_{y,max}(v, 0.8), 0.4g\right) \tag{42}$$

where $a_{y,max}(v, 0.8)$ denotes the maximum attainable lateral acceleration under the adopted dry-road model. This limit is derived from the simulation model and does not represent a validated capacity of a production EPS motor.

All controllers operate at a sampling period $T_c = 1$ ms. The lead–lag compensator is discretized using the Tustin method, while the PI/PID integrator employs forward Euler integration with back-calculation anti-windup. The integral and filtered derivative states are updated as

$$u_I[k+1] = u_I[k] + \backslash T_c \left(K_i e_T[k] + K_{aw}\left(T_{a,lim}[k] - T_{a,cmd}[k]\right)\right) \tag{43}$$

$$d[k] = a_d d[k-1] + b_d\left(T_s[k] - T_s[k-1]\right) \tag{44}$$

$$a_d = \frac{\tau_d}{\tau_d + T_c}, \quad b_d = \frac{K_d}{\tau_d + T_c} \tag{45}$$

where $T_{a,lim}$ denotes the assist-torque command after application of the common motor-block saturation limit. The difference between the limited and unlimited commands provides the back-calculation signal used to prevent integrator windup. The PI controller uses d[k] = 0, whereas PID uses the derivative state in Eq. (44). The anti-windup gain is:

$$K_{aw} = \frac{K_i}{K_p} \tag{46}$$

Sensor noise, quantization, and update periods are modeled according to Table 3, with resolutions taken from the sensor data sheets [14–16]. The noise standard deviation equals one resolution step for each sensor, except for vehicle speed, for which 0.38 km/h is adopted [17]. The 10 ms vehicle-signal update period is an implementation assumption.

Table 3. Sensor and actuator settings.

Signal or component	Resolution or setting	Update period
Steering torque T_s	0.01 N·m	1 ms
Steering-wheel angle θ_1	0.1°	1 ms
Column angular velocity $\dot{\theta}_2$	4°/s	1 ms
Vehicle speed v	1 km/h	10 ms
Yaw rate γ	0.1°/s	10 ms
Lateral acceleration a_y	0.01 g	10 ms
Controller sampling	1 ms	1 ms
Motor bandwidth	100 Hz	Continuous-time model
Computation delay	1 ms	-

The same five noise realizations (random-number seeds 99999 to 99995) are used for all controllers. The scoring results are reported as the mean and the sample standard deviation over the five realizations; the time histories in Figs. 8–10 show the realization with seed 99999. The motor current loop and electrical power constraints are not modeled. The initial 2 s of each simulation are excluded from performance evaluation.

4.2. Controller Calibration and Selection

The 117 s CALIB scenario contains steady cornering, sinusoidal steering, small lateral-acceleration corrections, and friction transitions. Steady-cornering tests are performed at 20, 60, and 100 km/h. The targets are 0.1, 0.15, and 0.18 g at 20 km/h, and 0.1, 0.2, and 0.3 g at 60

and 100 km/h. Each steady target is approached over 1 s and then held for 3 s, with the final second used for evaluation.

The calibration metric includes nine steady-cornering windows, three sinusoidal-steering windows at 0.5 Hz, three small-correction windows involving ±0.03 g target changes, and two friction-transition windows. At 60 km/h and a target of 0.3 g, road friction falls from 0.8 to 0.6 at 71 s and returns to 0.8 at 75 s. Two additional 0.4 g windows are retained for reporting but excluded from the calibration objective.

Torque-tracking accuracy and high-frequency assist-torque content are quantified by

$$R = \left(\frac{100}{4}\right) \sum_{q\in\{\text{static,sine,small,mu}\}} \backslash \left(\frac{RMS_q(e_T)}{mean_q\left(|T_{ref}|\right)}\right) \tag{47}$$

$$S = RMS\left(T_a - F_5(T_a)\right) \tag{48}$$

The metric R is the average normalized RMS tracking error across the four evaluation groups, expressed in percent. The metric S is computed as the RMS difference between the applied assist torque and its first-order low-pass-filtered counterpart, using a cutoff frequency of 5 Hz. It quantifies high-frequency assist-torque content, which may arise from both sensor-noise amplification and rapid control action. For steady-cornering windows, the signed relative error is calculated as:

$$e_{rel} = 100\frac{\text{sgn}\left(T_{ref}\right) mean(e_T)}{mean\left(|T_{ref}|\right)} \tag{49}$$

RMS error is used for steering-reversal windows, where signed relative errors may be misleading.

Candidate controllers must satisfy two admissibility requirements. The maximum absolute steady-state relative error across the nine calibration windows must not exceed 5%. Additionally, each configuration must remain stable under a doubled motor gain and an extra 1 ms computation delay, with the corresponding R value no greater than twice its nominal value. These are specified perturbation tests rather than a general guarantee of robust stability.

The final configuration is selected by minimizing

$$J = R \cdot S \tag{50}$$

among the admissible designs. To preserve the 0.2–2 Hz steering-response band, the lag stage of the assist-map compensator and the first-order derivative filters of PID are constrained to have a magnitude response no lower than −3 dB at 2 Hz. The PI controller has no corresponding filtering constraint.

4.3. Independent Evaluation Scenarios

After calibration, the controllers are tested on five independent scenarios. The conditions and principal evaluation measures are summarized in Table 4.

Table 4. Calibration and independent evaluation scenarios.

Scenario	Duration	Main operating conditions	Evaluation
CALIB	117 s	20, 60, 100 km/h; steady corners, 0.5 Hz steering and ±0.03 g corrections; μ = 0.8 → 0.6 → 0.8	Controller selection
DRY	129 s	Steady cornering at 20–100 km/h; μ = 0.8	Maximum steady relative error
SLIP	63 s	Low-friction patch, friction recovery and μ = 0.8 → 0.3 during cornering	Steady and transient torque errors
WEAVE	61 s	Slalom and sinusoidal steering under dry and low-friction conditions	RMS error, noise, reverse assist
RAMP	308.6 s	Vehicle-speed changes during steady cornering; μ = 0.8	Signed relative error
EDGE	87 s	High lateral acceleration of 0.35–0.4 g and sinusoidal steering	Tracking near operating limits

DRY evaluates tracking accuracy at operating speeds not fully represented by the calibration windows. SLIP tests the response to changing tire–road friction, whereas WEAVE examines continuous steering and steering-direction reversals. RAMP and EDGE provide additional diagnostic assessments rather than contributing to the calibration score.

Reverse-assist percentage is defined as the number of evaluated samples for which $T_a T_s$ is negative and $|T_s|$ is at least 0.5 N·m, divided by the total number of evaluated samples.

All three controllers are evaluated using identical steering inputs, speed profiles, friction conditions, and sensor-noise sequences. Their comparative results are presented in Section 5.

5. Results and Discussion

5.1. Calibration Performance and Tracking–Noise Trade-off

The CALIB parameter search evaluated 61 assist-map configurations, 453 PI configurations, and 824 PID configurations. Of the evaluated configurations, 56 assist-map, 149 PI, and 435 PID designs passed the steady-state accuracy gate. The stability-perturbation and filtering constraints were then applied to determine the admissible configurations used in the Pareto comparison. Fig. 7 shows the tracking metric R against the high-frequency assist-torque metric S for the evaluated designs and their admissible Pareto fronts.

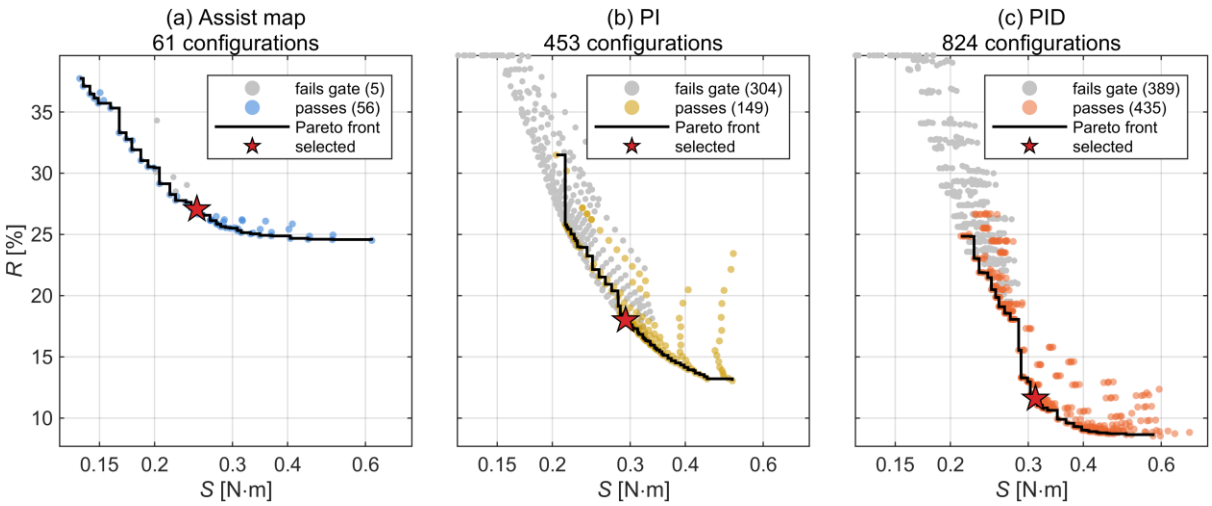


Fig. 7. Tracking–noise trade-off on CALIB: (a) assist map, (b) PI, (c) PID.

The evaluated assist-map and PID Pareto fronts share a high-frequency assist-torque range of approximately 0.21–0.62 N·m, allowing tracking errors to be compared at matched S values. At S = 0.25 N·m, the best admissible R values are 27.1% for the assist map and 20.5% for PID. At S = 0.30 N·m, these values are 25.5% and 13.0%, respectively. Thus, PID achieves lower tracking error at the two matched high-frequency assist-torque levels, while the assist map also provides admissible configurations at lower S values. Table 5 lists the selected designs. The tracking metric R decreases from 27.05% for the assist map to 18.00% for PI and 11.59% for PID. Over the same configurations, S increases from 0.249 to 0.293 and 0.312 N·m. All three designs satisfy the calibration accuracy gate and the specified gain- and delay-

perturbation tests. The reported perturbation ratios describe changes in R and should not be interpreted as classical gain or phase margins.

Table 5. Calibration performance of the selected controllers.

Controller	R (%)	S (N·m)	J = R·S	Maximum steady error (%)	Gain ×2 ratio	Delay +1 ms ratio
Assist map	27.05	0.249	6.74	3.17	0.78	1.00
PI	18.00	0.293	5.27	3.78	0.58	1.02
PID	11.59	0.312	3.61	2.33	0.40	1.01

The cost J is used solely to select a representative tracking–high-frequency-torque compromise within each controller family; it is not intended as a validated measure of steering feel.

5.2. Steering Torque Tracking under Dry-Road Conditions

The DRY scenario evaluates tracking performance under the calibration road condition across a wider set of steady-cornering operating points. Fig. 8 compares the steering torque response of the assist map, PI, and PID controllers.

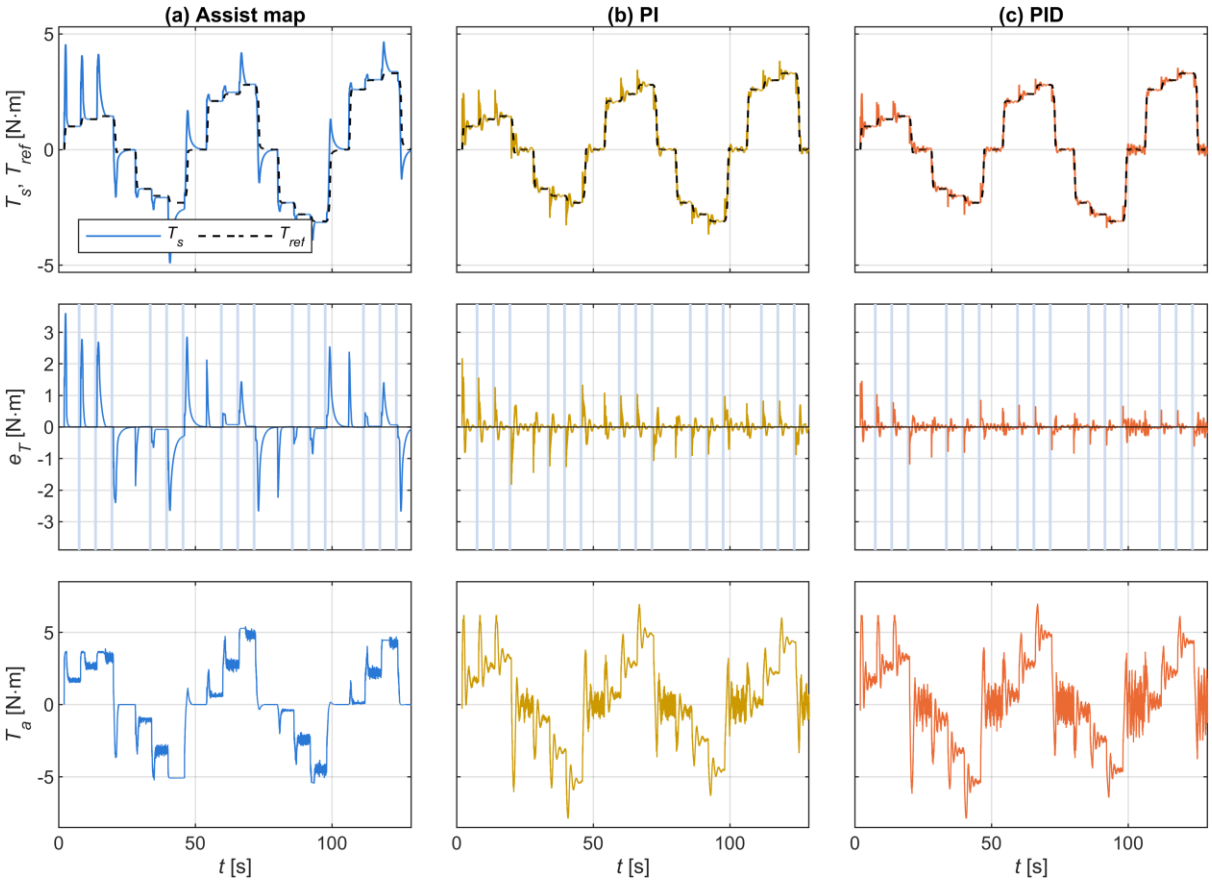


Fig. 8. DRY scenario: (a) assist map, (b) PI, (c) PID.

In the DRY scenario, the assist map reaches a maximum absolute steady-state error of 12.99% at 40 km/h and 0.3 g. At this point, measured steering torque exceeds the reference by approximately 0.30 N·m. In the 25–45 km/h and 0.25–0.3 g operating region, the unconstrained assist characteristic requires a local slope above the selected limit of 6. The slope-constrained map consequently provides less assistance than the ideal dry-road calibration characteristic, leaving the measured steering torque above its reference.

Across the DRY evaluation windows, the maximum absolute steady-state errors are 2.24% for PI and 1.77% for PID (standard deviations 0.41% and 0.39%). The difference between the two feedback controllers is within these standard deviations. The feedback controllers therefore maintain smaller worst-case errors across the evaluated operating points.

5.3. Torque Tracking under Varying Road Friction

The main objective of the proposed framework is assessed using the SLIP scenario. Fig. 9 compares the responses of the three controllers during low-friction disturbances and friction transitions.

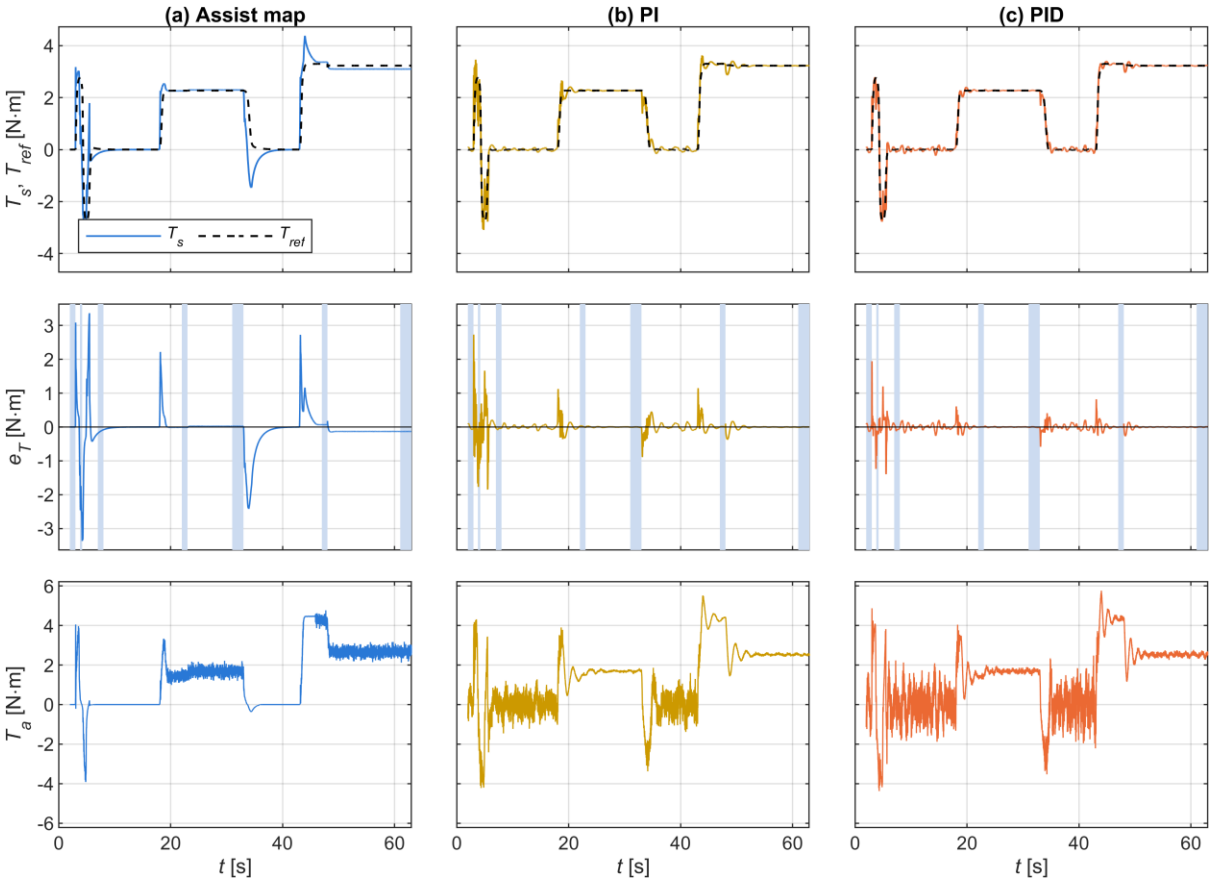


Fig. 9. SLIP scenario: (a) assist map, (b) PI, (c) PID.

In the SLIP scenario, road friction decreases from μ = 0.8 to 0.3 during a cornering maneuver at 100 km/h with a target lateral acceleration of 0.3 g. After the transition, the assist map retains a steady relative tracking error of −4.07%, compared with −0.08% for PI and −0.06% for PID (standard deviation 0.07% and 0.08%). The two feedback controllers are therefore both close to zero and cannot be distinguished from each other. The negative map error indicates that the measured steering torque is below the reference.

During the separate 0.4 s low-friction patch, the RMS tracking error is 2.30 N·m for the assist map, 0.85 N·m for PI, and 0.35 N·m for PID. In a friction-recovery event at 60 km/h and a target of 0.15 g, the corresponding steady relative errors are +1.01%, −0.15%, and −0.14%. Thus, the two integral-feedback controllers reduce the reported tracking errors for both friction reduction and recovery. These results apply to the simulated steering inputs and vehicle response; they do not isolate the map-slope effect from changes in lateral acceleration and reference torque during the transitions.

5.4. Dynamic Steering Performance and Practical Limitations

The remaining scenarios test continuous steering, changing vehicle speed, and operation at high lateral acceleration. Fig. 10 compares the tracking errors of the three controllers across DRY, RAMP, SLIP, WEAVE, and EDGE. Table 6 reports whole-case RMS tracking error, high-frequency assist-torque content, and reverse-assist percentage as mean ± standard deviation over the five noise realizations. Reverse assistance is defined here by the sign relationship between applied motor torque and measured steering torque. It does not, by itself, indicate closed-loop instability, actuator saturation, or a loss of steering control.

PID has the lowest whole-case RMS tracking error in all five scenarios. In WEAVE, for example, the RMS error decreases from 1.327 N·m for the assist map to 0.376 N·m for PI and 0.215 N·m for PID. The corresponding S values increase from 0.160 to 0.332 and 0.407 N·m. The tracking improvement is therefore accompanied by greater high-frequency assist-torque activity during continuous steering.

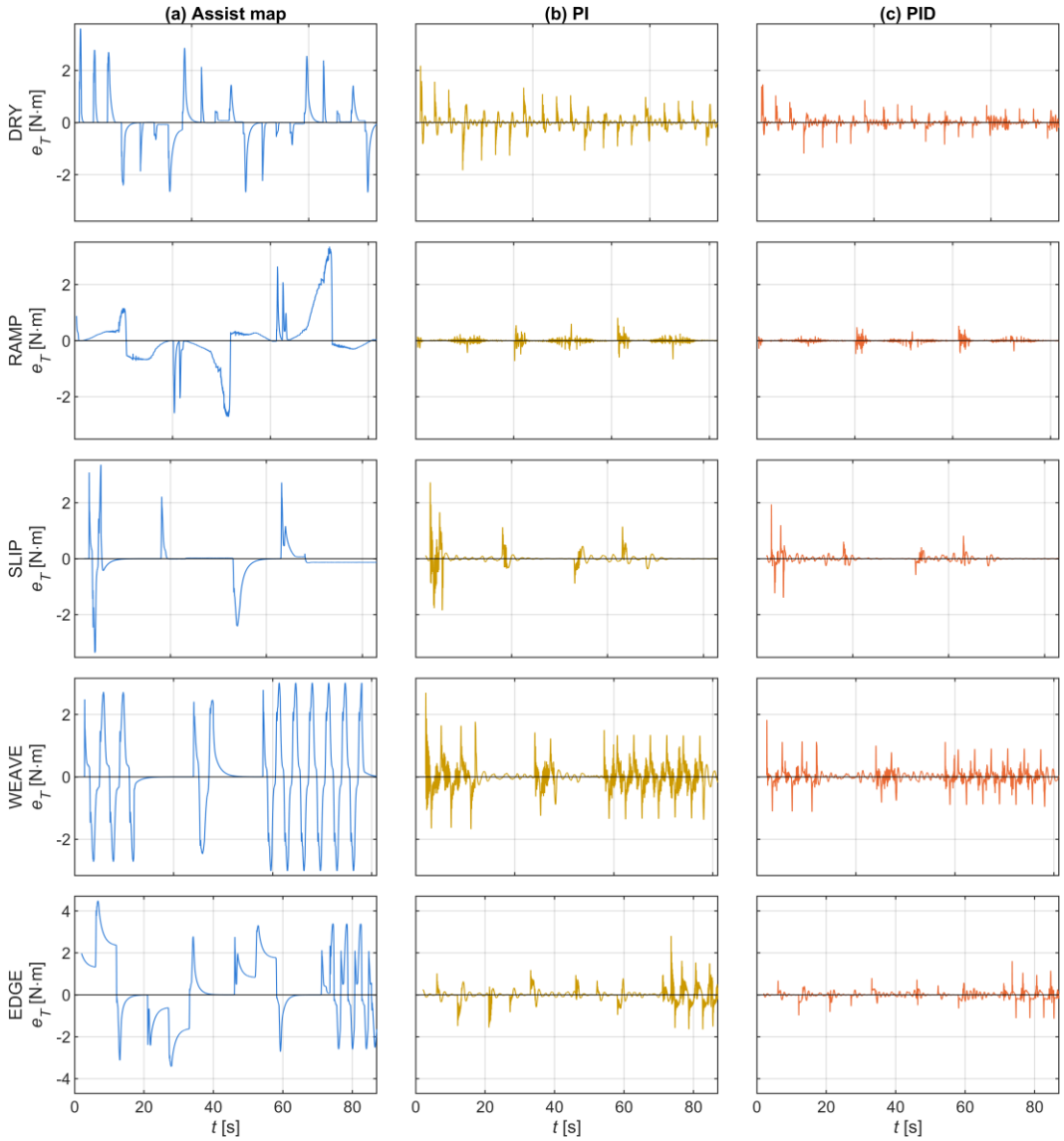


Fig. 10. Tracking error in the five scenarios: (a) assist map, (b) PI, (c) PID.

Table 6. Whole-case performance of the investigated controllers (mean ± standard deviation over five noise realizations).

Scenario	Controller	RMS tracking error (N·m)	S (N·m)	Reverse assist (%)
DRY	Assist map	0.689 ± 0.000	0.149 ± 0.000	0.14 ± 0.02
DRY	PI	0.244 ± 0.002	0.195 ± 0.002	6.04 ± 0.15
DRY	PID	0.154 ± 0.001	0.255 ± 0.002	5.83 ± 0.14
SLIP	Assist map	0.538 ± 0.000	0.163 ± 0.000	0.04 ± 0.01
SLIP	PI	0.216 ± 0.002	0.236 ± 0.004	3.37 ± 0.09
SLIP	PID	0.128 ± 0.004	0.304 ± 0.004	3.49 ± 0.07
WEAVE	Assist map	1.327 ± 0.000	0.160 ± 0.000	0.05 ± 0.01
WEAVE	PI	0.376 ± 0.004	0.332 ± 0.004	23.47 ± 0.05
WEAVE	PID	0.215 ± 0.003	0.407 ± 0.003	25.50 ± 0.17
RAMP	Assist map	0.835 ± 0.000	0.169 ± 0.001	0.00 ± 0.00
RAMP	PI	0.070 ± 0.001	0.088 ± 0.001	5.89 ± 0.04
RAMP	PID	0.046 ± 0.002	0.127 ± 0.001	5.84 ± 0.04
EDGE	Assist map	1.534 ± 0.000	0.130 ± 0.000	0.02 ± 0.00
EDGE	PI	0.300 ± 0.003	0.255 ± 0.004	6.97 ± 0.24
EDGE	PID	0.163 ± 0.002	0.321 ± 0.003	6.67 ± 0.09

The whole-case RMS values in Table 6 describe the overall tracking performance of each scenario. To examine persistent errors at individual operating points, signed relative errors are evaluated separately over steady-state windows. These two measures are not directly interchangeable.

The RAMP case reveals a limitation of the assist map that is less apparent from the whole-case RMS values. During decreasing-speed cornering, its signed steady relative error reaches +15.2%, +31.4%, and +46.7% at the three evaluated lateral-acceleration levels. During increasing-speed cornering, the corresponding errors are −20.2%, −7.4%, and −5.8%. PI and PID remain within approximately 0.5% in these windows.

In EDGE, the largest assist-map steady relative errors occur at 40 km/h, reaching +56.7% at 0.35 g and +99.7% at 0.4 g. The corresponding PI errors are −0.4% and −0.5%, while the PID errors are −0.1% and −0.2%. At these high lateral accelerations, the reference

torque increases only slightly while the road-reaction torque continues to rise. The resulting natural assistance characteristic requires a steep local slope. Limiting that slope to 6 prevents the assist map from delivering the calibrated torque, leaving the measured steering torque substantially above its reference. The EDGE results are treated as diagnostic tests near the limits of the adopted model. Their quantitative interpretation remains conditional on the validity of the small-angle assumptions at the evaluated operating points.

The additional RAMP and EDGE cases also indicate that the assist map is more sensitive to speed variation and high lateral acceleration, whereas the PI/PID controllers remain closer to the desired torque reference within the simulated operating range.

5.5. Discussion

The assist-map errors arise from two mechanisms. The first is the local slope limit. A road-reaction torque disturbance is attenuated by $1/(1 + s_m)$, so a finite slope leaves a residual error. The same constraint prevents the map from reaching its unconstrained calibration characteristic where the required slope is high, as observed in DRY and EDGE.

The second mechanism is column Coulomb friction during continuous steering motion. In RAMP, the friction torque approaches ±2 N·m, with its sign determined by the direction of column motion. Its contribution to the steering-torque balance changes sign between decreasing- and increasing-speed maneuvers. The assist map only partially compensates for this disturbance, whereas integral feedback reduces the persistent torque error. These mechanisms explain why the map shows different error patterns in friction-transition, steady-cornering, and speed-ramp tests.

The reverse-assist results limit the interpretation of the tracking improvements. A possible cause is the delayed reversal of the integral contribution relative to the measured steering torque, but the individual events have not been analyzed to confirm this mechanism. The comparison is also affected by the 0.3 N·m dead zone in the assist map, which is absent from PI and PID. Reverse-assist percentage alone does not establish whether the driver would perceive the response as undesirable; that question requires driver-in-the-loop testing.

The comparison also reflects a difference in available feedback information: PI and PID use lateral acceleration to construct the torque reference, whereas the conventional assist map uses only measured steering torque and vehicle speed. The reported differences therefore characterize the complete controller architectures, not the isolated effect of PI or PID action.

Several limitations remain. The steering–vehicle model has not been validated against measurements from a physical vehicle, and the predicted road-reaction torque depends on the assumed pneumatic-trail parameters. The driver is represented by a prescribed steering angle, while motor current-loop dynamics and power constraints are omitted. The desired torque is taken from one published reference characteristic. Each controller was calibrated with a single noise realization, whereas the scoring cases are averaged over five realizations. In addition, the three controller searches used different numbers of parameter candidates, and the assist map includes a torque dead zone that is absent from PI and PID. These differences should be considered when interpreting the tracking and reverse-assist comparisons.

6. Conclusions

This study compared a conventional assist map with PI and PID steering-torque feedback for a column EPS system under changing tire–road friction. The torque reference was constructed from vehicle speed and lateral acceleration, without explicit friction estimation.

The analysis identified a steady-state limitation of the assist map: changes in road-reaction torque are attenuated according to the local map slope, which is constrained by stability and noise requirements. In the nonlinear simulations, a friction reduction from $\mu = 0.8$ to 0.3 produced a steady relative tracking error of −4.07% for the assist map, compared with −0.08% for PI and −0.06% for PID. The PID controller also reduced the calibration tracking metric from 27.05% to 11.59%.

These improvements came with increased high-frequency assist-torque content and substantial reverse assistance during continuous steering. The results establish a tracking advantage for the investigated PI/PID configurations, but not an improvement in subjective steering feel. Further work should extend the parameter-sensitivity analysis beyond pneumatic trail, examine reverse-assist events in greater detail, and validate the controller responses experimentally.

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Conflict of Interest

The authors have no conflict of interest to declare.

Author contributions

Tran Minh Tan: modeling, simulation, data analysis, writing – original draft. Phung Thanh Huy: supervision, review and editing. All authors have approved the final article.

Declaration of Generative AI

During the preparation of this work, the author(s) used Claude (Anthropic) in order to assist with simulation code, figure generation, data checking, and language editing. After using this tool, the author(s) reviewed and edited the content as needed and take full responsibility for the content of the published article.

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